Muneeb Arshad

Potsdam, NY

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EDUCATION

CLARKSON UNIVERSITY Potsdam, NY, USA

PhD Candidate, Electrical and Computer Engineering (Expected Graduation: December 2026) January 2022 – Present

Astronautics and Robotics (ASTRO) Laboratory

Advisor: Dr. Michael Bazzocchi

Coursework: Deep Learning, Space Robotics, Orbital Mechanics and Dynamics, Advanced Computer-Aided Design (CAD), Software System Architecture

Research Focus:

- Modeled a soft robotic gripper for autonomous space debris capture and removal
- Developed a grasp pose estimation method for identifying suitable capture configurations for irregularly shaped space debris
- Developed a deep reinforcement learning-based control strategy to minimize perturbations in free-floating space manipulators during the pre-capture phase of space debris capture
- Developed deep reinforcement learning approaches for autonomous capture and detumbling of space debris using free-floating robotic systems

CLARKSON UNIVERSITY Potsdam, NY, USA

Master of Science in Electrical and Computer Engineering January 2022 – April 2025

Astronautics and Robotics (ASTRO) Laboratory

Advisor: Dr. Michael Bazzocchi

Research Focus:

- Investigated space debris removal approaches and close-proximity capture techniques for autonomous on-orbit servicing and debris mitigation applications
- Evaluated advanced terrestrial gripping technologies, including soft grippers, for their potential adaptation to space-based robotic manipulation
- Analyzed robotic control methods and teleoperation strategies to support reliable and effective space debris capture operations

NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY Islamabad, Pakistan

Master of Science in Electrical Engineering (Control System) September 2019 – December 2021

Coursework: Advanced Digital Signal Processing, Machine Learning, Linear Control Systems, Stochastic Systems, Advanced Power Electronics, Image Processing for Intelligent Systems, Artificial Intelligence, Mobile Robotics

Thesis: Analysis and Classification of EEG Signals

Research Focus:

- Investigated EEG signal processing, feature extraction, and classification using deep learning algorithms

COMSATS UNIVERSITY Islamabad, Pakistan

Bachelor of Science in Electrical and Computer Engineering September 2015 – April 2019

Final Year Project: Design of Auto tracker for Solar Panels

Project Focus:

- Designed a slew drive mechanism and control module to enable automatic solar tracking for improved energy harvesting

RESEARCH PUBLICATIONS

JOURNAL PUBLICATIONS

- M. Arshad, M.C.F. Bazzocchi, F. Hussain, “Emerging Strategies in Close Proximity Operations for Space Debris Removal: A Review” Acta Astronautica, Vol. 228, pp. 996-1022, 2025.
- M. Arshad, M. C. F. Bazzocchi, “Perturbation and Disturbance Minimization of a Free-Floating Space Manipulator Through Deep Reinforcement Learning Control” Aerospace Science and Technology, Minor revisions.
- M. Arshad, M. C. F. Bazzocchi, “Deep Reinforcement Learning Control for Detumbling a Non-Cooperative Space Target” under review.
- M. Arshad, M. C. F. Bazzocchi, “Deep Reinforcement Learning for Stable Grasping of a Non-Cooperative Space Target Using a Soft Robotic End-Effector” in preparation.
- M. Arshad, M. C. F. Bazzocchi, “Detection and Estimation suitable Grasp Pose for Space Debris Capture” in preparation.

CONFERENCE PUBLICATIONS

- M. Arshad, M.C.F. Bazzocchi, “Collision Avoidance and Disturbance Minimization through Deep Reinforcement Learning Control of a Free-Floating Space Manipulator,” 75th International Astronautical Congress (IAC), Milan, Italy, Oct. 14-18, 2024.

EXPERIENCE

YORK UNIVERSITY Toronto, Canada
International Visiting Research Trainee January 2024 – January 2025

- Investigated deep reinforcement learning-based control strategies for collision avoidance and disturbance minimization in free-floating space manipulators.
- Developed a robust control framework using deep reinforcement learning to enable smooth and safe pre-capture operations of free-floating space manipulators.
- Investigated methods for detecting and estimating suitable grasp poses for space debris capture

MAANZ-AI Islamabad, Pakistan
Associate C++ Developer January 2021 – December 2021

- Contributed to the development of a self-driving car simulation platform
- Worked with Scala and Elasticsearch for database development and data management
- Developed and integrated REST APIs for software communication and system integration

ENGINEERING DEVELOPMENT BOARD Islamabad, Pakistan
Intern March 2020 – December 2020

- Contributed to the development of a policy for power equipment and the energy sector
- Conducted trade analysis of energy goods between Pakistan and international markets
- Analyzed import and export trends of energy goods to support policy and market assessments

NATIONAL INSTITUTE OF ELECTRONICS Islamabad, Pakistan
Intern July 2018 – August 2018

- Contributed to the development and testing of custom PCBs
- Contributed to the development of energy-efficient SSL/LED lighting products

PEER REVIEWER FOR JOURNALS

Reviewed 20 research manuscripts as a peer reviewer for leading Elsevier space journals, including:

- Aerospace Science and Technology (15)
- Acta Astronautica (3)
- Advances in Space Research (2)

PRESENTATIONS

- Muneeb Arshad, *"Collision Avoidance and Disturbance Minimization through Deep Reinforcement Learning Control of a Free-Floating Space Manipulator"*, 75th International Astronautical Congress (IAC), Milan, Italy, Oct. 14-18, 2024.

HONORS AND AWARDS

- Tuition fellowship, Clarkson University, USA
- International Visiting Research Opportunity, York University, Canada
- Punjab Educational Endowment Fund Scholarship
- Punjab Group of Colleges Merit Scholarship

CERTIFICATIONS

- Robotics in Space, International Space Alliance (ISA).
- Artificial Intelligence, Presidential Initiative for Artificial Intelligence and Computing (PIAIC).
- Systems Tool Kit (STK) Level 2: Master Certification, Ansys
- Systems Tool Kit (STK) Level 1 Certification, Ansys
- AI for Everyone, Coursera

SOFTWARE SKILLS

- MATLAB, Xilinx, ORCAD, Proteus, Atmel Studio, Visual Studio, SolidWorks, ANSYS, CREO, AutoCAD, MeshLab, 3D Zephyr, Fusion 360

PROGRAMMING SKILLS

- Python, C++, C, Assembly language, JAVA, Scala

OPERATING SYSTEMS

- Windows, Linux

EXTRA CURRICULAR ACTIVITIES

- Participated in Inter-Departmental Quiz Competition, COMSATS University Islamabad
- Participated in Robotics Competition, COMSATS University Islamabad
- Member, Pakistani Student Association, Clarkson University

WORKSHOPS

- National Workshop on Application of Robotics Technology for Industrial Development - NUST
- Arduino, COMSATS University
- PCB Fabrication, COMSATS University
- National Police Summit and Innovation Expo, Islamabad

ADDITIONAL INFORMATION

- Languages: Urdu (Native), English (Advanced)
- Interests: Horse Ridding, Swimming, Cricket, Volleyball